

KESHAV SINGLA

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Backend software engineer with production experience in FastAPI, Node.js and Python, plus a hands-on security background spanning secure coding, OWASP-aligned code review, and forensic/compliance tooling.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C++, SQL

Backend & Systems: FastAPI, Spring Boot, Node.js/Express, REST APIs, Microservices, Docker, Nginx

Security: Secure coding, OWASP Top 10, code review, PII isolation & access control, OCR/forgery detection pipelines

Data & Tools: PostgreSQL, SQLite, SQLAlchemy, Supabase, Git, Linux, unit/integration testing (pytest)

AI/ML: PyTorch, OpenCV, RAG pipelines, model evaluation, on-device inference

EXPERIENCE

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AI/ML Engineer Intern

Remote

Jun 2026 – Present

- Own the on-device vitals pipeline for a production health module: built rPPG-based signal processing (Python, OpenCV, NumPy/SciPy) from raw camera frames to heart-rate estimates, verified at 70.9 BPM against real hardware with 265 passing automated tests.
- Wrote unit and integration tests as a core deliverable of the module rather than an afterthought, and documented engineering decisions in structured build plans and implementation briefs for founder/PM review.
- Reviewed third-party dependencies against a strict open-source licence policy (Apache-2.0/MIT/BSD-3 only) before they reached production, flagging and resolving licence risks before shipping.
- Collaborate asynchronously with a distributed team, producing daily written checkouts and structured technical documents (task trackers, model comparisons) as the primary mode of engineering communication.

EvoDoc HealthTech

Backend Developer Intern

Chennai, India

Feb 2026 – Mar 2026

- Built and deployed a FastAPI backend for a clinical document-processing platform, implementing PII isolation, role-based access control, and secure parsing pipelines used in production.
- Designed a misinformation-verification service (RAG-based fact-checking over a streaming data source and a public fact-check API), reviewing code for correctness and security before merge.

SECURITY-FOCUSED PROJECTS

AutoJustice AI NEXUS — FastAPI, PostgreSQL, Docker, OCR

Nov 2025 – Present

- Built a backend platform automating cybercrime FIR intake with OCR extraction, image-forgery detection, and rule-based compliance checks against Section 65B, the DPDP Act, and the BNS; containerized with Docker behind Nginx.
- Built a professional network of 26+ contacts across Indian law enforcement and cybersecurity through direct outreach, including validation from a Maharashtra Police officer who escalated the platform to Mumbai's Cyber DCP within 24 hours.

ObscuraAI — PyTorch, Gradient Reversal, CelebA

Ongoing

- Co-authoring a research project auditing adversarial demographic-attribute leakage in face-recognition embeddings, targeting an arXiv preprint as first author.

Automated Network Security Scanner — Python, Nmap

- Built a scanning tool for vulnerability and open-port detection, applying OWASP-style checks to flag common misconfigurations.

CERTIFICATIONS

- Deloitte Cyber Job Simulation (Forage)
- OPSWAT Critical Infrastructure Protection

EDUCATION

SRM Institute of Science and Technology

B.Tech, Computer Science & Engineering — CGPA: 7.76/10

Chennai, India

Expected Oct 2028

- Relevant coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Cryptography & Network Security, System Design
- Technical Team Lead, DSA SRMIST; Campus Crew Member, HackerRank & CODENEX Club